

Pooling Intraclass Correlation Coefficients After Multiple Imputation in Cluster-Randomized Trials

Missing data are common in health, education, and social science research. The problem can be especially challenging in cluster-randomized trials (CRTs). In a CRT, entire groups, such as schools, hospitals, clinics, or communities, are randomly assigned to treatment conditions instead of assigning each person separately. The people within the same cluster may have similar outcomes because they share the same environment, staff, policies, or resources. Therefore, the observations cannot be treated as completely independent.

Multiple imputation (MI) is one method for handling missing data. Instead of replacing each missing value with one fixed value, such as the overall mean, MI creates several completed datasets containing different plausible values for the missing observations. The statistical model is then fitted separately to each completed dataset. Finally, the results are combined using Rubin's rules. Rubin's rules produce a pooled parameter estimate and standard error by accounting for both ordinary sampling variation and variation between the imputed datasets. In clustered data, the imputation method should also account for the cluster structure so that important differences between clusters are not removed during imputation.

This thesis will focus on pooling an intraclass correlation coefficient (ICC) after MI. In a simple random-intercept mixed model, the ICC is the proportion of the total outcome variance that is due to differences between clusters. For example, an ICC of 0.20 means that 20% of the total variation in the outcome is associated with differences between clusters, while the remaining 80% occurs between individuals within clusters. The ICC is important because it describes the strength of clustering and affects standard errors, effective sample size, and the design and analysis of CRTs.

A difficulty occurs when Rubin's rules are applied directly to ICC estimates. Standard scalar Rubin pooling works best when the complete-data estimator has an approximately normal sampling distribution on the scale being pooled. An estimated ICC may not meet this approximation— ICC estimates are bounded, may be skewed, and may accumulate at zero when variance estimates are constrained to be nonnegative. Raw-scale Rubin pooling is still mathematically possible, but its standard errors and confidence intervals may be inaccurate or may extend outside the meaningful ICC range.

The main goal of this project is to investigate whether transforming the ICC before applying Rubin's rules produces more reliable pooled estimates and confidence intervals. The raw ICC will be included as a comparison method. Possible transformations will include Fisher's z transformation (commonly used for correlation coefficients), the logit transformation, and the arcsine square-root transformation. Log-based transformations and the Freeman-Tukey double-arcsine transformation may also be considered. Notably, there is no obvious transformation that is guaranteed to work best for an ICC.

For this thesis, we can use a simulation study based on a continuous-outcome random-intercept mixed model. Data will be generated with a known true ICC, after which missing values will be introduced and handled using MI. A mixed model will be fitted to each imputed dataset. For each candidate method, the ICC estimate and its variance will be transformed, pooled using Rubin's rules, and then returned to the original ICC scale. The methods will be compared using bias, empirical variance, mean squared error, confidence-interval coverage, and other evaluation criteria. Because the true ICC is known in a simulation, it will be possible to determine which method most accurately recovers it. A real dataset could then be used as a practical illustration, although the true ICC would not be known in that application.

Potential Dryad datasets for the applied portion include the following:

1. A cluster-randomized trial involving 2,461 caregiver-child pairs in rural Peru that compared an AI-supported parenting chatbot and traditional home visits, with continuous child-development scores as outcomes: <https://doi.org/10.5061/dryad.nzs7h4551>
2. A cluster-randomized hypertension trial involving 757 patients from 32 community health centers in Ghana. Systolic blood pressure provides a useful continuous outcome: <https://doi.org/10.5061/dryad.16c9m51>
3. A South African cluster-randomized home-visiting trial with continuous outcomes including height-for-age scores, stunting, EEG measures, and reaction time: <https://doi.org/10.5061/dryad.qnk98sfm2>

This topic interests me because it connects a practical missing-data problem with an important feature of clustered research. MI and Rubin's rules are widely used, but bounded and skewed parameters such as the ICC may require more careful treatment. The project could provide useful guidance about when raw pooling is adequate and when a transformation gives more accurate estimates and confidence intervals.

References

- Van Buuren, S. Flexible Imputation of Missing Data, Chapter 5: Multiple-imputation workflow and parameter pooling.
<https://stefvanbuuren.name/fimd/workflow.html>
<https://stefvanbuuren.name/fimd/sec-pooling.html>
- Eldridge, S. M., Ukoumunne, O. C., and Carlin, J. B. (2009). The Intra-Cluster Correlation Coefficient in Cluster Randomized Trials: A Review of Definitions.
<https://doi.org/10.1111/j.1751-5823.2009.00092.x>
- Murray, D. M., Taljaard, M., Turner, E. L., and George, S. M. (2020). Essential Ingredients and Innovations in the Design and Analysis of Group-Randomized Trials.
<https://doi.org/10.1146/annurev-publhealth-040119-094027>
- Lin, L., and Xu, C. (2020). Arcsine-based transformations for meta-analysis of proportions: Pros, cons, and alternatives.
<https://doi.org/10.1002/hsr2.178>